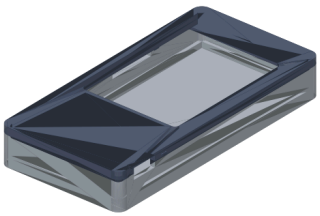
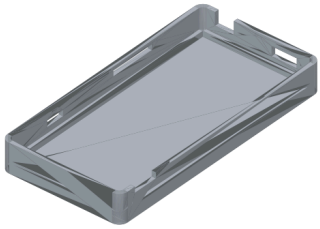
Pod case, Rev A

Two 3D-printed parts: a **lower tray** that holds the perfboard, and an **upper bezel** that frames the screen and snaps onto the tray. No screws show anywhere: the stack is clamped between the tray's rails and the bezel, and the bezel is held by six snap bumps on its lip. Outer size **57.2 × 114.2 × 22.7 mm**, 2 mm walls, 4 mm corner radius.

Assembled



Lower tray, from above (USB end bottom-right)



Upper bezel, turned over (inside)



Upper bezel, from above



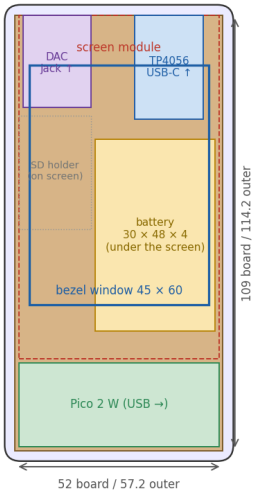
Openings

Opening	Wall	Size	For
Headphone jack	top end, left	9 × 9, notch from z 10.1 up through the tray rim	DAC's 3.5 mm jack
USB-C charging	top end, right	12 × 6	TP4056 module's socket
Pico micro-USB	right, near the bottom	12 × 6.8, closed by the bezel	flashing firmware
microSD	left, middle	22 × 3.5	the screen module's card holder
Power switch	left, near the bottom	9 × 4	slide switch lever

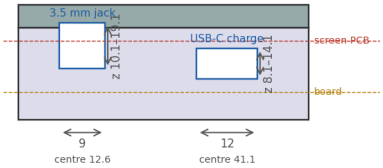
All openings are 1–2 mm bigger than the part so the plug overmoulds fit. Positions come from the perfboard layout in *docs/perfboard_build.pdf*; the sheet that follows gives every centre and height.

Dimensions (mm)

Plan view, looking into the tray (board coordinates)



Top end wall (jack + USB-C), seen from outside



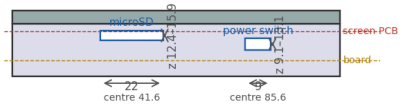
Right wall (Pico USB), seen from outside



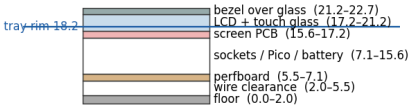
Key numbers

```
Outer: 57.2 × 114.2 × 22.7 mm, corner radius 4
Walls 2.0, floor 2.0, bezel top 1.2 over the glass
Board clearance 0.6 per side; lip fit 0.25 per side
Tray height 18.2; bezel lip drops 3 into the tray
Board rests on 1.6 mm rails, 3.5 above the floor
Bezel window 45 × 60, centre (26, 66.5) board coords
Six snap bumps on the lip / pockets in the tray
No external screws: snap-fit; stack is clamped
between the rails and the bezel's underside.
```

.eft wall (SD + switch), seen from outside (mirror of plan)



Stack (section), mm from the outside of the floor



Where the new parts go

- **Battery (403048, 4 × 30 × 48):** between the perfboard and the screen, in the clear zone **x 20–50, y 30–78** (board coordinates, Pico end = bottom). That zone is under the screen, right of the SD holder, above the screen's socket row and below the TP4056. Stick it down with a strip of foam tape; 4 mm thick in an 8.5 mm gap leaves room for the screen's solder stubs.
- **TP4056 (26 × 17):** on the board top at the top-right corner, USB-C socket poking 1 mm past the top edge, foam tape underneath. Wires: OUT+ → switch → 1N5819 → Pico VSYS; OUT− → GND; battery to B+ / B−.
- **Power switch (right-angle slide):** mounted on the board's left edge so its lever sticks out through the left-wall slot; or glue it into the slot from inside and run two wires.
- **100 nF** across R2 as in the build guide; the **diode** in the power path as above.

Print settings

- PLA or PETG, 0.2 mm layers, 3 walls, 20 % infill. Tray prints open side up, bezel prints face down (window side on the bed). No supports needed.
- Tolerance built in: 0.6 mm around the board, 0.25 mm on the lip. If the bezel is tight, sand the lip; if loose, a drop of glue on two bumps.

Before printing in plastic: three checks

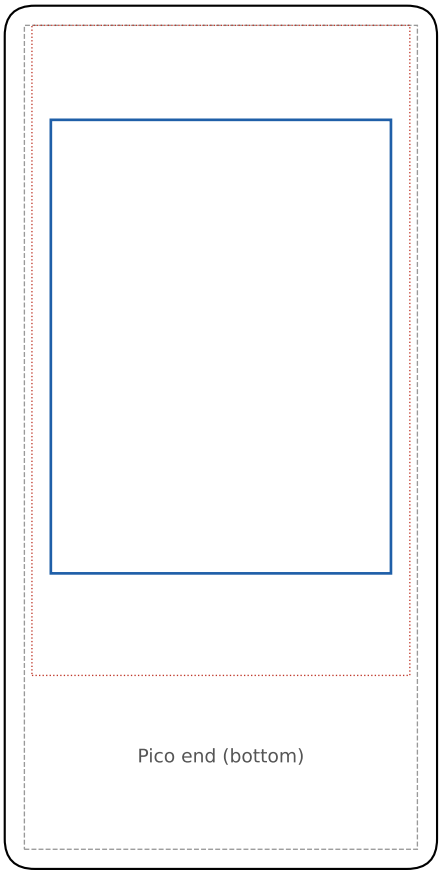
1. **Print the 1:1 bezel template** (last page, 100 % scale) and lay it on the device. The blue window must frame the touch panel with 1–2 mm of glass under the bezel on each side. If it's off, tell me by how much and which way; it's a two-number change.
2. **Measure the jack height:** from the perfboard's top surface to the centre of the jack barrel. The model assumes ~7.5 mm with a 9 mm tall slot; if yours is outside 4–12 mm, say so.
3. **Measure the SD card slot height** on the screen's underside, if you want the card accessible without opening the case.

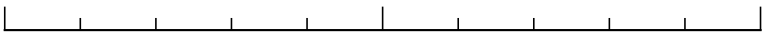
Assembly

1. Fit the switch and TP4056 wires, battery in its zone, all taped down. Kapton over the solder side.
2. Drop the board into the tray, jack end into the jack notch first, then the Pico USB into its notch.
3. Press the bezel on until the six bumps click. To open: a fingernail or card in the seam at the Pico end.

Files: *pod_tray.stl* / *.step*, *pod_bezel.stl* / *.step*, *pod_case_assembled.step*, and the parametric source *pod_case.py*: every number above is a named parameter at the top of that file. The STEP files open directly in Fusion 360.

Pod bezel, 1:1. Print at 100%. Lay it on the screen with the dashed board outline on the perfboard:
the blue window must frame the touch panel with 1–2 mm of glass hidden under the bezel on every side.





100 mm check line